ACEITEKA UACE BIO 1 PROPOSED GUIDE BY TR. SSEMATIMBA GODFREY

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ITEM 1

(a) Analyse the findings to explain how the Mega-Boost supplement affected the various cellular parameters leading to the observed symptoms in the collapsed athletes.

The **ATP concentration in the muscle cells of athletes using Mega-Boost was 18.6 mmol kg^{-1} muscle compared with 5.2 mmol kg^{-1} in normal athletes**. **This is linked to the severe muscle fatigue and collapse experienced by the athletes despite apparently having a higher ATP concentration.** ATP is the immediate energy currency used to power muscle contraction, active transport and other cellular processes. During prolonged exercise, ATP is continuously broken down to ADP and must be rapidly resynthesised through cellular respiration. The supplement was reported to produce NADH substances directly inside the mitochondria and slow carbohydrate breakdown. Excess NADH can interfere with the normal balance of oxidised and reduced electron carriers, while high ATP concentration can also exert feedback inhibition on respiratory pathways. Therefore, the high measured ATP concentration does not necessarily indicate that the muscles could continuously generate and utilise ATP at the rate required during a marathon. When ATP demand increased greatly during prolonged exercise, the disrupted respiratory pathways could fail to maintain adequate energy turnover, resulting in severe fatigue and eventual collapse.

The **pyruvate concentration decreased from 3.3 mmol L^{-1} in normal athletes to 1.2 mmol L^{-1} in athletes using Mega-Boost**. **This is linked to the severe fatigue experienced during long-distance races because the supplement was reported to slow down carbohydrate breakdown.** Pyruvate is the end product of glycolysis, which occurs in the cytoplasm and breaks glucose into smaller molecules while producing ATP and reduced NAD. Pyruvate normally enters the mitochondrion and is converted into acetyl-CoA during the link reaction before entering the Krebs cycle. When carbohydrate breakdown is slowed, less pyruvate is produced and consequently less substrate is available for the link reaction and Krebs cycle. This reduces the supply of hydrogen/electrons to the electron transport chain and limits the ability of the muscles to sustain aerobic ATP production. During a long-distance race, where muscles require continuous energy, this reduced carbohydrate contribution can cause the energy supply to fall below the energy demand, resulting in exhaustion and collapse.

The **lactic acid concentration increased dramatically from 1.5 mmol L^{-1} in normal athletes to 14.2 mmol L^{-1} in athletes using Mega-Boost**. **This is directly linked to the severe muscle fatigue and collapse observed during the long-distance races.** When aerobic respiration cannot provide sufficient ATP to meet the energy demand of actively contracting muscles, the muscles increasingly rely on anaerobic respiration. During anaerobic respiration, pyruvate is converted to lactate, allowing NAD to be regenerated so that glycolysis can continue. The very high lactate concentration therefore indicates substantial reliance on anaerobic metabolism. Accumulation of lactate and associated hydrogen ions lowers the pH of muscle tissue, interfering with enzyme activity and reducing the efficiency of actin-myosin interactions during contraction. Consequently, muscle contraction becomes

less effective, producing weakness and severe fatigue and eventually contributing to the athletes' collapse.

The **rate of lipid breakdown was very high in athletes using Mega-Boost compared with a moderate rate in normal athletes**. **This is linked to the collapse of the athletes because they appear to have shifted heavily towards lipid metabolism while carbohydrate breakdown was being suppressed.** Lipids are hydrolysed into fatty acids and glycerol, with fatty acids undergoing β-oxidation in the mitochondria to produce acetyl-CoA, reduced NAD and reduced FAD. These products enter aerobic respiratory pathways to generate ATP. However, fatty-acid oxidation is relatively slow and requires substantial oxygen. During a long-distance race, muscles require ATP at a rapidly changing rate, particularly when exercise intensity increases. Excessive dependence on lipid oxidation may therefore fail to provide ATP rapidly enough to match the immediate energy requirements of contracting muscles. This imbalance between energy demand and energy supply contributes to the severe fatigue observed in the athletes.

The **mitochondrial cristae became swollen following prolonged use of Mega-Boost**. **This structural change is linked to the athletes' inability to sustain efficient aerobic respiration and their subsequent collapse during long-distance races.** The cristae are folds of the inner mitochondrial membrane and provide a large surface area containing electron transport chain proteins and ATP synthase. Electron carriers transfer electrons through the electron transport chain, releasing energy that is used to pump hydrogen ions across the inner mitochondrial membrane. This establishes an electrochemical gradient that drives ATP synthesis through ATP synthase. Swelling and disruption of the cristae can interfere with the organisation and functioning of these respiratory proteins, reducing the efficiency of electron transport and oxidative phosphorylation. Consequently, muscle cells become less capable of producing ATP aerobically when energy demand is high. During prolonged exercise, this impairment becomes particularly important because the muscles require a continuous supply of ATP, ultimately contributing to severe fatigue and collapse.

The **combination of very high lipid breakdown, reduced pyruvate concentration and greatly increased lactic acid concentration** demonstrates a major disturbance in the normal balance between energy pathways. **This is linked to the athletes' inability to maintain endurance during the long-distance races.** Normally, carbohydrates and fats are metabolised in a coordinated manner to supply ATP according to the intensity and duration of exercise. In the affected athletes, carbohydrate breakdown was suppressed, forcing greater reliance on lipid metabolism, while the high lactate concentration indicates that anaerobic respiration was also becoming increasingly important. At the same time, mitochondrial cristae were swollen, compromising aerobic ATP production. Therefore, the supplement disrupted both the **availability of respiratory substrates and the efficiency of the organelle responsible for aerobic energy production**, causing the energy supply to become inadequate for sustained muscle contraction and resulting in fatigue and collapse.

(b) Evaluate the effectiveness of the interventions made by the medical team.

1. Administering 100% oxygen gas to the collapsed athletes during recovery

Administering oxygen can be effective during recovery because it increases the availability of oxygen for aerobic respiration. Oxygen acts as the final electron acceptor in the electron transport chain, allowing oxidative phosphorylation to continue. Increased oxygen availability can therefore support ATP production in functional mitochondria. It can also assist recovery from the oxygen deficit created when muscles relied heavily on anaerobic respiration. Additional oxygen allows accumulated lactate to be oxidised and helps restore ATP and other energy reserves following strenuous exercise. Therefore, oxygen administration can help the collapsed athletes recover from the immediate effects of prolonged exercise.

However, oxygen administration cannot directly correct the underlying mitochondrial and metabolic disruption caused by Mega-Boost. The supplement produced NADH substances directly inside the mitochondria and caused swelling of the mitochondrial cristae. If the inner mitochondrial membrane and its electron transport components are structurally or functionally disrupted, increasing oxygen concentration alone cannot restore their normal organisation or completely re-establish efficient oxidative phosphorylation. Furthermore, very high oxygen availability can increase the formation of reactive oxygen species, which may cause oxidative damage to cellular proteins, lipids and membranes.

Therefore, oxygen is an appropriate supportive intervention during recovery, particularly where oxygen demand is high, but it should not be regarded as a treatment for the underlying effects of Mega-Boost. The supplement should be discontinued and the metabolic disturbance addressed.

2. Feeding the athletes a high-glucose syrup immediately after they collapsed

Providing glucose can help replenish carbohydrate supplies after prolonged exercise. Glucose enters glycolysis and can provide respiratory substrate for ATP production. It can also contribute to the restoration of glycogen stores in the muscles and liver after prolonged exercise. Since long-distance exercise uses substantial amounts of stored carbohydrate, replacing these stores can support recovery and future exercise performance.

However, a high-glucose syrup is unlikely to immediately correct the main metabolic disturbance in these athletes because Mega-Boost was reported to slow carbohydrate breakdown, and the pyruvate concentration was already reduced from 3.3 to 1.2 mmol L^{-1}. If glycolysis is inhibited, supplying more glucose does not necessarily result in a corresponding increase in pyruvate production. In addition, the athletes already had a very high ATP concentration of 18.6 mmol kg^{-1} compared with 5.2 mmol kg^{-1} in normal athletes. The immediate problem was therefore not simply a lack of glucose or ATP but disrupted regulation and utilisation of respiratory pathways. Excess glucose could be stored as glycogen or converted into other metabolites rather than immediately solving the impaired energy production.

Therefore, glucose supplementation may be useful for restoring depleted carbohydrate reserves during recovery, but it is not the most effective immediate treatment for the metabolic disturbance caused by Mega-Boost. It should be used as a supportive measure rather than the primary intervention.

3. Administering a drug that competitively inhibits fatty-acid oxidation before the race

Inhibiting fatty-acid oxidation could potentially reduce the excessive dependence on lipid metabolism observed in the affected athletes. The rate of lipid breakdown was very high compared with the moderate rate in normal athletes. By competitively inhibiting fatty-acid oxidation, the drug would reduce the utilisation of fatty acids as respiratory substrates and could encourage greater reliance on carbohydrate metabolism. Carbohydrates can provide ATP more rapidly than fats and are relatively more efficient when rapid ATP production is required during intense exercise. Therefore, reducing excessive lipid oxidation could potentially improve the balance between energy pathways.

However, this intervention could actually worsen the athletes' condition because carbohydrate breakdown was already being slowed by Mega-Boost. Long-distance athletes require both carbohydrate and lipid metabolism to maintain ATP production over prolonged periods. If fatty-acid oxidation is inhibited while carbohydrate utilisation is simultaneously impaired, the muscles may have insufficient usable respiratory substrate. In addition, fatty-acid oxidation is an important source of ATP during prolonged exercise, particularly after glycogen stores begin to decline. Completely or excessively suppressing this pathway could therefore accelerate energy depletion rather than prevent fatigue.

Therefore, competitively inhibiting fatty-acid oxidation is not an appropriate stand-alone solution. The better approach would be to discontinue Mega-Boost and restore normal coordination between carbohydrate and lipid metabolism rather than blocking one of the body's major aerobic energy pathways.

Of the three interventions, **administering oxygen is the most appropriate immediate supportive measure during recovery**, because it increases oxygen availability for aerobic respiration and assists the removal of the metabolic consequences of anaerobic respiration. **High-glucose syrup can be useful for replenishing carbohydrate stores**, but its immediate effectiveness is limited because the supplement has already disrupted carbohydrate breakdown. **Inhibiting fatty-acid oxidation is the least suitable intervention**, because it removes an important source of energy at a time when carbohydrate metabolism is already impaired.

The most effective overall management would therefore be to **discontinue Mega-Boost**, provide appropriate supportive oxygen during recovery, restore normal nutritional and metabolic balance, and monitor the athletes until mitochondrial and respiratory function return towards normal.

ITEM 2

(a) Analyse how the conditions subjected to the peppermint plant groups influence productivity in the plants.

The two peppermint groups were subjected to different **light durations and ABA concentrations**, while both experienced moderate temperature. Group A received **8 hours of light and moderate ABA**, whereas Group B received only **6 hours of light and high ABA**. These differences were associated with major differences in plant performance: Group A had a **high photosynthetic rate, moderate wilting and flowering**, while Group B had a **low photosynthetic rate, severe wilting and no flowering**. Therefore, the conditions imposed on

the plants influenced productivity mainly through their effects on photosynthesis, water relations and reproductive development.

The **longer light exposure in Group A, 8 hours compared with 6 hours in Group B, was associated with a high photosynthetic rate in Group A compared with a low rate in Group B**. This difference in photosynthesis is linked to the greater productivity of Group A and the failure of Group B to perform well. Light provides the energy required for the light-dependent reactions of photosynthesis. Chlorophyll absorbs light energy, which is used to produce ATP and reduced NADP. These products are then used in the Calvin cycle to drive carbon fixation and the production of carbohydrates. Therefore, the longer period of light exposure in Group A provided more opportunity for the plants to capture light energy and produce assimilates. These carbohydrates can subsequently be used in respiration to provide energy and as raw materials for growth and development. The shorter light period experienced by Group B therefore limited the energy available for photosynthesis, contributing to its low photosynthetic rate and reduced productivity.

The **ABA concentration was moderate in Group A but high in Group B, and this was accompanied by moderate wilting in Group A compared with severe wilting in Group B**. This relationship links the high ABA condition in Group B to the poorer water status and reduced productivity of those plants. ABA is a plant hormone associated with responses to water stress. When a plant experiences water shortage, ABA concentration increases and promotes stomatal closure. Closing the stomata reduces the loss of water through transpiration and therefore helps the plant conserve water. However, stomatal closure also reduces the diffusion of carbon dioxide into the leaf. Since carbon dioxide is required for carbon fixation during the Calvin cycle, reduced CO_2 entry lowers the rate of photosynthesis. Thus, although high ABA helps the plant conserve water, the high ABA condition in Group B also contributes to reduced photosynthesis and therefore reduced productivity.

The **severe wilting in Group B compared with moderate wilting in Group A** indicates that the environmental conditions subjected to Group B produced greater water stress. This is linked to the **low photosynthetic rate and poor productivity** observed in Group B. Water is necessary to maintain cell turgidity. When water is lost faster than it can be replaced, cells lose turgor pressure and leaves become limp, producing wilting. Under water stress, ABA concentration increases, causing stomatal closure. Although this reduces further water loss, it also restricts CO_2 uptake and therefore reduces photosynthetic carbon fixation. Severe water stress can consequently produce a cycle in which water shortage increases ABA, ABA promotes stomatal closure, stomatal closure restricts CO_2 entry, and reduced CO_2 availability lowers photosynthesis and carbohydrate production. This helps explain why Group B had low photosynthesis and poor productivity.

The **Group A plants flowered, whereas Group B plants did not flower**. This difference is linked to the **longer light exposure, moderate ABA concentration and better physiological condition of Group A compared with Group B**. Flowering requires adequate energy and assimilates because the plant must develop reproductive structures. The high photosynthetic rate of Group A would increase carbohydrate production, providing energy and carbon compounds required for reproductive development. In contrast, the shorter light period and high ABA concentration in Group B were associated with low photosynthesis and severe wilting. Consequently, fewer assimilates were available for reproductive development, while the stressed plants were more likely to prioritise survival and water conservation rather

than reproduction. This explains the absence of flowering in Group B and therefore its lower reproductive productivity.

The fact that **both groups were maintained at moderate temperature** indicates that temperature was relatively similar between the two treatments and therefore does not explain the major difference in productivity between the groups. Instead, the major differences were the **duration of light exposure and ABA concentration**. Group A, which received longer light exposure and moderate ABA, maintained high photosynthesis, moderate wilting and flowering. Group B, which received shorter light exposure and high ABA, experienced low photosynthesis, severe wilting and no flowering. This comparison demonstrates that environmental conditions and hormonal regulation interacted to influence the physiological performance of the peppermint plants.

Overall, the conditions subjected to the two groups influenced productivity by affecting **energy capture, water balance, carbon dioxide uptake and reproductive development**. Group A received more light and had moderate ABA, enabling relatively effective photosynthesis and flowering. Group B received less light and high ABA, which was associated with severe water stress, stomatal closure, reduced CO_2 uptake, low photosynthesis and failure to flower. Therefore, the conditions imposed on Group B created physiological limitations that reduced both vegetative and reproductive productivity.

(b) Evaluate the effectiveness of the interventions proposed by the scientists and give strategic guidance.

1. Routine watering of the plants, especially during hot days

Routine watering would be **an effective intervention for improving the productivity of the peppermint plants**, particularly those experiencing severe wilting. The severe wilting observed in Group B indicates poor water status. Supplying adequate water allows roots to absorb water and transport it through the xylem to leaves and other tissues. Water maintains cell turgidity, allowing leaves to remain firm and physiologically functional. Adequate water availability would also reduce water stress and consequently reduce the need for high ABA levels.

However, excessive watering would not necessarily improve productivity. If too much water accumulates in the soil, the air spaces become filled with water and oxygen availability around the roots decreases. Root cells require oxygen for aerobic respiration and ATP production. Reduced oxygen availability can therefore interfere with active transport of mineral ions and other root processes, potentially reducing plant growth. Excessive irrigation may also encourage water-related diseases.

Therefore, routine watering is an effective intervention, but it should be **controlled and based on the water requirements of the plants rather than simply watering excessively**. Watering should be increased particularly during hot or dry periods when water loss through transpiration is high.

Strategic guidance

The farmer should use **efficient irrigation practices**, such as watering when soil moisture becomes low, rather than maintaining continuously saturated soil. Mulching could also be used to reduce evaporation from the soil surface and conserve water. Maintaining adequate soil moisture would reduce water stress, reduce excessive ABA responses and help maintain stomatal function and photosynthesis.

2. Increasing soil salinity by adding ash in the gardens

Increasing soil salinity by adding ash would generally be **an ineffective and potentially harmful intervention for improving peppermint productivity**. Although ash may contain mineral substances that can contribute to soil fertility, deliberately increasing soil salinity can make it more difficult for plant roots to obtain water. A highly concentrated soil solution has a lower water potential, reducing the water potential gradient between the soil and root cells. Consequently, water uptake by the roots can decrease.

This is particularly problematic because the plants are already showing severe wilting. Increasing soil salinity could intensify water stress, even when water is present in the soil. Greater water stress would stimulate ABA production, which promotes stomatal closure. Reduced stomatal opening would restrict carbon dioxide entry into the leaves and consequently reduce photosynthesis. Therefore, increasing salinity could worsen the exact problems already observed in Group B: **severe wilting and low photosynthetic rate**.

However, ash can sometimes contain mineral nutrients that may benefit plant growth when applied appropriately. If a soil is deficient in certain nutrients, carefully managed application of suitable amendments could improve nutrient availability and plant development. The potential benefit, therefore, comes from correcting nutrient deficiencies rather than from deliberately increasing salinity.

Therefore, adding ash specifically to increase soil salinity should **not be recommended** as a strategy for improving peppermint productivity. If ash is considered as a soil amendment, its composition and application rate should first be assessed to ensure that it does not create harmful salinity.

Strategic guidance

The farmer should prioritise **maintaining suitable soil water potential and fertility rather than increasing salinity**. Soil testing should be carried out before applying amendments. If nutrients are deficient, appropriate fertilisers or soil amendments should be applied at recommended rates. This would improve plant nutrition without creating additional osmotic stress.

Between the two proposed interventions, **routine and controlled watering is clearly more effective than increasing soil salinity**. Watering directly addresses the severe wilting observed in the plants and can improve water availability, maintain cell turgidity and reduce water-stress responses. Reduced water stress would also reduce excessive ABA-mediated stomatal closure, allowing greater CO_2 entry and supporting photosynthesis.

In contrast, **increasing soil salinity is unsuitable**, especially under conditions where the plants are already experiencing severe wilting. High salinity can reduce water uptake through osmotic effects, increase water stress and stimulate ABA responses. This can cause greater stomatal closure, lower CO_2 uptake and further reduction in photosynthesis.

However, **watering alone may not completely solve the productivity problem**, because Group B also received only **6 hours of light compared with 8 hours in Group A**. The low photosynthetic rate was therefore associated with both reduced light exposure and high ABA. Strategic management should consequently combine **adequate water supply with sufficient light exposure, appropriate soil fertility and careful management of plant stress**.

Final judgement

The recommended strategy is controlled watering during periods of high water demand, combined with water-conservation practices such as mulching and appropriate irrigation scheduling. Soil salinity should not deliberately be increased. In addition, the farmer should ensure that peppermint plants receive adequate light and should monitor soil moisture and plant stress. This integrated approach is more likely to restore photosynthesis, support flowering and ultimately improve peppermint productivity sustainably.

ITEM 3

(a) Explain how David's parameters varied from those of a healthy student and how this led to David's presentation.

David's **ADH concentration was much lower, at 0.8 pg/mL compared with 4.5 pg/mL in the healthy student**. This difference is linked to David's **profound dehydration** after the six-hour outdoor music festival and alcohol consumption. ADH normally acts on the kidney's collecting ducts, increasing their permeability to water by promoting the insertion of aquaporin water channels into their membranes. This increases water reabsorption from the filtrate back into the blood. Alcohol suppresses ADH release, so less water is reabsorbed and a larger volume of dilute urine is produced. Consequently, David would lose more water from the body, causing dehydration and contributing to the fluid imbalance observed on admission.

David's **cardiac output was also lower, at 3.2 L/min compared with 5.1 L/min in the healthy student**. This is linked to his **dehydrated condition and reduced physical performance following the festival**. Loss of water reduces blood volume, thereby reducing venous return to the heart. Reduced venous return lowers the amount of blood filling the ventricles during diastole and consequently reduces stroke volume. Since cardiac output depends on heart rate and stroke volume, a reduction in stroke volume can result in reduced cardiac output. Therefore, David's severe dehydration contributed to the lower amount of blood being pumped around his body each minute, which could reduce the delivery of oxygen and nutrients to actively respiring tissues.

The **sensory hair cells in David's cochlea were 35% damaged, whereas the healthy student had 0% damage**. This difference is directly linked to David's complaint of a **constant ringing in his ears, or tinnitus**, after attending a music festival characterised by extreme noise levels. Sensory hair cells in the cochlea are responsible for converting mechanical vibrations caused by sound into electrical impulses that can be transmitted to the

auditory nerve. Exposure to extremely loud sound produces excessive movement and stimulation of these delicate hair cells, which can damage their structures. Damaged hair cells may produce abnormal sensory signals that the brain interprets as ringing or buzzing even when no external sound is present. Therefore, the 35% cochlear hair-cell damage provides a biological explanation for David's persistent ringing in the ears.

The **endolymph fluid dynamics in David's semicircular canals were described as highly turbulent with altered viscosity, whereas the healthy student had normal flow**. This difference is linked to David's **difficulty walking in a straight line**. The semicircular canals contain endolymph and sensory receptors that detect rotational movements of the head and contribute to maintaining balance. When the head moves, movement of the endolymph causes displacement of the sensory structures, generating nerve impulses that provide information about the direction and extent of head movement. If the normal flow and viscosity of the endolymph are disturbed, the movement of the fluid may no longer accurately correspond to head movements. This can produce abnormal signals to the brain regarding body orientation and movement, resulting in dizziness, impaired coordination and difficulty maintaining balance. Therefore, the altered endolymph dynamics help explain why David could not walk in a straight line.

The **low ADH concentration and reduced cardiac output are also physiologically connected to David's dehydration**. His ADH concentration of **0.8 pg/mL compared with 4.5 pg/mL** would reduce renal water conservation, while his prolonged dancing and exposure to an outdoor environment would increase water loss through sweating. The resulting reduction in body water would decrease blood volume, which explains why cardiac output fell from **5.1 L/min to 3.2 L/min**. Reduced circulation can further affect exercise tolerance because less oxygenated blood is delivered to tissues, particularly when the body is already recovering from prolonged physical activity.

Overall, David's presentation resulted from several interacting disturbances. **Alcohol-associated suppression of ADH contributed to excessive water loss and profound dehydration; dehydration reduced blood volume and cardiac output; extreme noise exposure damaged cochlear sensory hair cells, producing tinnitus; and altered endolymph dynamics in the semicircular canals interfered with balance, causing difficulty walking in a straight line.** The data therefore demonstrate how changes in physiological parameters were directly associated with the symptoms presented by David.

(b) Evaluate the medical team's proposal in helping David and support the attending doctor's advice.

The medical team proposed **intravenous isotonic saline, oral rehydration salts or plain distilled water, together with placement in a quiet rehabilitation environment**. These interventions can help address different aspects of David's condition, but their effectiveness differs depending on the physiological problem being treated.

Intravenous isotonic saline

Intravenous isotonic saline would be effective in treating David's profound dehydration, particularly because his cardiac output had fallen to 3.2 L/min compared with 5.1 L/min in the healthy student. Isotonic saline has an osmotic concentration close to that of body fluids, so it can increase extracellular fluid volume without causing major movement of water into or out of body cells. When administered directly into a vein, it rapidly increases blood volume. This can improve venous return, increase ventricular filling and help restore cardiac output.

However, intravenous saline does not correct the underlying suppression of ADH caused by alcohol, nor does it repair the 35% damage to cochlear hair cells or directly restore abnormal endolymph dynamics. Excessive administration can also cause fluid overload if not carefully controlled.

Therefore, intravenous isotonic saline is an effective immediate treatment for severe dehydration, but it should be medically monitored and should be combined with measures that address the underlying cause.

Oral rehydration salts

Oral rehydration salts would also be effective in restoring David's water and electrolyte balance once he is sufficiently stable and able to drink. The solution contains water, glucose and electrolytes. Glucose and sodium are absorbed together in the small intestine, and water follows the absorbed solutes by osmosis. This promotes efficient absorption of both water and electrolytes. This is particularly useful because David had been dancing for six hours and would have lost water and salts through sweating.

However, oral rehydration salts may not be sufficient as the initial treatment for profound dehydration if David is unable to drink adequately or requires rapid restoration of circulating volume. They also cannot reverse the cochlear damage or the altered endolymph dynamics.

Therefore, oral rehydration salts are effective for continued rehydration when David can safely drink, but intravenous fluid replacement is more appropriate when dehydration is severe.

Plain distilled water

Plain water can help replace some of the water lost through dehydration, and therefore it can contribute to recovery. However, **distilled water contains very few dissolved electrolytes**, whereas prolonged sweating results in the loss of both water and mineral ions such as sodium and chloride. Replacing large amounts of water without replacing electrolytes can dilute the concentration of electrolytes in body fluids and disturb osmotic balance.

Therefore, plain distilled water is less effective than oral rehydration salts for significant dehydration following prolonged physical activity. David would benefit more from an appropriate electrolyte-containing fluid, particularly during the initial recovery period.

Quiet rehabilitation environment

Placing David in a **quiet rehabilitation environment would be effective in preventing further exposure of his damaged cochlear hair cells to intense sound**. The table shows that 35% of his sensory hair cells were damaged compared with none in the healthy student. Removing him from loud noise reduces further mechanical stimulation of the cochlear structures and gives any reversibly affected auditory cells an opportunity to recover.

However, a quiet environment cannot necessarily repair sensory hair cells that have already suffered permanent structural damage. Therefore, it may prevent further deterioration and support recovery, but it cannot be considered a complete treatment for his tinnitus or cochlear damage.

Therefore, the quiet environment is an appropriate supportive intervention, but David would also require continued assessment of his hearing and auditory symptoms.

Supporting the attending doctor's advice to quit alcohol

The attending doctor's advice that **David should quit alcohol is strongly supported by the physiological evidence in the case**. His ADH concentration was only **0.8 pg/mL compared with 4.5 pg/mL in the healthy student**. Alcohol suppresses ADH secretion, reducing the kidney's ability to conserve water. With less ADH, the collecting ducts become less permeable to water, so less water is reabsorbed into the blood and more water is lost in urine. Continued alcohol consumption would therefore increase the risk of repeated dehydration.

This is particularly important because David already presented with **profound dehydration and a reduced cardiac output of 3.2 L/min**. If he continues consuming alcohol, repeated suppression of ADH could continue to promote water loss, making it difficult to maintain normal blood volume. Reduced blood volume can again reduce venous return, stroke volume and cardiac output.

However, simply stopping alcohol will not immediately correct the dehydration, low cardiac output or existing cochlear damage. David still requires appropriate immediate medical management, including fluid replacement and protection from further loud noise.

Nevertheless, quitting alcohol is essential for long-term recovery because it removes an important cause of his excessive water loss and reduces the likelihood of recurrent dehydration. It also prevents continued alcohol-related physiological disturbances while the body recovers.

Overall judgement

The most appropriate immediate approach is **controlled fluid and electrolyte replacement together with rest in a quiet environment**. For profound dehydration, intravenous isotonic saline is likely to provide the fastest restoration of circulating volume, while oral rehydration salts are useful once David can safely drink. Plain distilled water is less suitable for substantial dehydration because it does not adequately replace lost electrolytes. A quiet environment is important for preventing further noise-related damage, although it cannot guarantee repair of the damaged cochlear hair cells.

The attending doctor's advice to **quit alcohol should be strongly supported**. The markedly reduced ADH concentration provides physiological evidence that alcohol-related suppression of water conservation contributed to David's dehydration. Therefore, **fluid replacement addresses the immediate effects, while quitting alcohol addresses an important underlying cause and helps prevent recurrence**.

ITEM 4

(a) Analyse the data and account for the differences in the physiological parameters assessed in the three players.

The three players used different strategies before, during and after the football match, and this was reflected in their **synaptic transmission efficiency, reaction time, heart rate and skin temperature**. Player 1 completed a structured warm-up, Player 2 received a local anaesthetic after a muscle injury and continued playing, while Player 3 received ice-cold treatment immediately after injury and returned to play shortly afterwards. These different interventions therefore produced different effects on nervous coordination, circulation and temperature regulation.

Player 1 had high synaptic transmission efficiency and the fastest reaction time of 160 ms, compared with reduced synaptic transmission efficiency and a reaction time of 260 ms in Player 2 and moderate efficiency with 210 ms in Player 3. The superior nervous performance of Player 1 is linked to his structured warm-up before playing. A warm-up progressively increases muscle and tissue temperature and prepares the nervous system for physical activity. Increased temperature within physiological limits can enhance the speed of nerve impulse transmission and improve the efficiency of synaptic processes. At the neuromuscular junction, this supports rapid communication between motor neurons and muscle fibres, allowing muscles to respond quickly to motor commands. Consequently, Player 1 had the fastest reaction time, which would favour rapid responses to movements of the ball and opponents during the match.

Player 2 had reduced synaptic transmission efficiency and the slowest reaction time of 260 ms, compared with Player 1's 160 ms. **This is linked to the local anaesthetic administered after his muscle injury and his decision to continue playing.** Local anaesthetics reduce the generation and propagation of nerve impulses by interfering with the movement of ions, particularly sodium ions, across neuronal membranes. This reduces the ability of sensory nerves to transmit pain and other information effectively. Although reducing pain may allow Player 2 to continue playing, reduced sensory signalling can interfere with rapid communication between the nervous system and muscles. This helps explain his reduced synaptic transmission efficiency and slower reaction time. His slower response could impair coordination and increase the likelihood of making mistakes or sustaining further injury during a high-speed game.

Player 3 had moderate synaptic transmission efficiency and a reaction time of 210 ms, which was better than Player 2 but slower than Player 1. **This is linked to the immediate application of ice-cold treatment following his muscle injury and his return to play shortly afterwards.** Cold temperatures reduce the rate of nerve impulse transmission because

they slow the movement of ions and the functioning of membrane proteins involved in action potentials. Cooling also reduces muscle enzyme activity and can temporarily reduce neuromuscular responsiveness. Therefore, although ice treatment may have helped control the immediate effects of injury, the reduced tissue temperature could have temporarily impaired nerve and muscle function, accounting for Player 3's intermediate reaction time.

Player 1 recorded the highest post-play heart rate of 150 beats per minute, compared with 138 beats per minute in Player 2 and 132 beats per minute in Player 3. This difference can be linked partly to the physiological state produced by their different interventions and their ability to sustain active muscular performance. During football, contracting muscles require increased supplies of oxygen and respiratory substrates. The sympathetic nervous system increases heart rate so that cardiac output rises and more oxygenated blood is delivered to active muscles. Player 1's effective warm-up prepared his cardiovascular system for exercise, allowing increased circulation during vigorous activity. His higher heart rate is therefore consistent with a strong cardiovascular response to exercise.

However, Player 2's heart rate was lower at 138 beats per minute despite continuing to play with a muscle injury, compared with Player 1's 150 beats per minute. **This may be linked to the effect of the local anaesthetic and the injured state of the player.** Because the anaesthetic reduces pain sensation, Player 2 may have been able to continue playing despite tissue damage. However, masking pain does not remove the underlying physiological injury. The reduced sensory feedback can interfere with normal protective responses and may allow the player to continue loading an injured muscle. His cardiovascular response may therefore not fully reflect the extent of the physical stress imposed on the injured body.

Player 3 recorded the lowest heart rate, at 132 beats per minute, together with the lowest skin temperature of 32.5°C. This is linked to the **ice-cold treatment applied immediately after injury**. Cold exposure causes vasoconstriction, reducing blood flow to the cooled region. It also reduces metabolic activity and heat transfer to the skin. Consequently, the skin temperature falls considerably. Reduced peripheral blood flow can also influence the cardiovascular response during the immediate recovery period. The combination of low skin temperature and moderate reaction time therefore reflects the physiological effects of cooling on circulation and nervous system function.

The skin temperature also differed considerably among the players: Player 1 had 38.2°C, Player 2 had 36.4°C and Player 3 had only 32.5°C. The high skin temperature in Player 1 is linked to the structured warm-up and continued muscular activity. Muscle contraction increases metabolic activity, and a substantial proportion of the chemical energy released during respiration is converted into heat. Increased blood flow to the skin allows heat to be transferred from deeper tissues to the body surface, where it can be lost to the surroundings. This helps explain Player 1's relatively high skin temperature after play.

In contrast, **Player 3's skin temperature of 32.5°C was much lower than Player 1's 38.2°C and Player 2's 36.4°C**. **This is directly linked to the ice-cold treatment applied after his injury.** Cold ice removes heat from tissues and causes vasoconstriction, reducing blood flow to the cooled area. Consequently, less warm blood reaches the skin and the temperature of the tissue falls. While this can be beneficial for reducing excessive swelling and inflammation following an acute injury, excessive or prolonged cooling can also slow nerve conduction and muscle function. This provides a physiological explanation for Player 3's moderate synaptic transmission efficiency and reaction time of 210 ms.

Overall, the data show that the three strategies produced different physiological outcomes. **Player 1's structured warm-up was associated with high synaptic efficiency, the fastest reaction time and the highest heart rate and skin temperature**, indicating effective preparation for physical activity. **Player 2's local anaesthetic was associated with reduced synaptic efficiency and the slowest reaction time**, suggesting that masking pain came with impairment of sensory and nervous responses. **Player 3's ice treatment produced the lowest skin temperature and an intermediate reaction time**, reflecting the effects of cold on circulation and nerve function. Therefore, the interventions affected performance by altering **nerve transmission, sensory feedback, cardiovascular responses and tissue temperature**.

(b) Evaluate each player's strategy and assess which one balances performance and physiological safety.

Player 1: Structured warm-up before play

Player 1's structured warm-up appears to be the **most effective strategy for supporting performance while maintaining physiological safety**. His **synaptic transmission efficiency was high and his reaction time was 160 ms**, the fastest of the three players. This indicates effective nervous-system preparation for the demands of football. A warm-up gradually increases muscle temperature and blood flow, improves the readiness of the nervous system and prepares muscles and joints for vigorous movement. This can improve coordination and reduce the likelihood of injury associated with suddenly subjecting cold muscles to intense activity.

However, Player 1 had the highest heart rate of 150 beats per minute and skin temperature of 38.2°C. This shows that his body was experiencing substantial cardiovascular and thermal demand after play. If exercise continues under hot conditions, continued heat production could increase the risk of excessive body temperature and dehydration. Therefore, a warm-up improves readiness but must be followed by appropriate hydration, rest and heat management.

Therefore, Player 1's strategy provides the best balance between performance and safety, provided that the warm-up is structured and excessive heat accumulation is prevented.

Player 2: Local anaesthetic after muscle injury and continued play

Player 2's strategy may provide a **short-term performance advantage because the local anaesthetic reduces pain sensation**, allowing him to continue participating despite his muscle injury. This could explain why he was able to remain in the match rather than stopping immediately because of pain.

However, the physiological disadvantages are significant. Player 2 had **reduced synaptic transmission efficiency and the slowest reaction time of 260 ms**. Local anaesthetics reduce

nerve impulse transmission by interfering with sodium ion movement across neuronal membranes. This reduces sensory signalling, meaning the nervous system receives less information from the injured area. Pain normally acts as a protective mechanism that causes a person to modify movement or stop an activity that could worsen an injury. By blocking this sensation, the anaesthetic can mask the severity of the injury while the underlying tissue damage remains.

The slow reaction time could also impair coordination and increase the risk of further injury during rapid football movements. Continued play may place additional mechanical stress on already damaged muscle fibres, potentially increasing the severity of the injury.

Therefore, although the local anaesthetic may temporarily improve the player's ability to participate, it does not provide a good balance between performance and physiological safety. The strategy should not be used simply to enable an injured player to continue competing. Proper assessment and recovery should take priority.

Player 3: Ice-cold treatment immediately after injury and early return to play

Player 3's strategy has **some important benefits immediately after an acute muscle injury**. Cold treatment causes vasoconstriction and reduces local blood flow and metabolic activity. This can help limit excessive swelling and reduce inflammatory responses in the injured area. Cooling can also reduce pain by slowing sensory nerve activity.

However, the data show a major physiological cost: Player 3 had a skin temperature of only 32.5°C, the lowest of the three players, together with moderate synaptic transmission efficiency and a reaction time of 210 ms. Cold temperatures slow nerve impulse transmission and reduce the rate of biochemical reactions involved in muscle contraction. Therefore, returning to play shortly after applying ice may expose the player to reduced neuromuscular responsiveness. This could impair coordination and increase the risk of further injury because the muscles and nervous system may not yet have returned to their optimal functional state.

Therefore, ice treatment can be effective as part of immediate injury management, but returning to play shortly afterwards is not the safest strategy. The player should first allow adequate recovery and ensure that normal muscle temperature, movement and neuromuscular function have returned before resuming competitive activity.

Overall assessment and strategic guidance

Of the three strategies, **Player 1's structured warm-up provides the best balance between performance and physiological safety**. He recorded the **highest synaptic transmission efficiency and fastest reaction time of 160 ms**, indicating good neuromuscular readiness. His higher heart rate of 150 bpm is consistent with vigorous exercise, while his skin temperature of 38.2°C indicates substantial heat production; therefore, hydration, rest and appropriate cooling should accompany the strategy.

Player 2's strategy is the least safe for an injured player. Although the local anaesthetic may temporarily suppress pain and allow participation, his reduced synaptic transmission

efficiency and slowest reaction time of 260 ms show that nervous-system performance was compromised. More importantly, masking pain may encourage continued loading of damaged tissue and increase the risk of worsening the injury.

Player 3's ice treatment is useful for managing the immediate effects of acute injury, but the decision to return to play shortly afterwards reduces its safety. His skin temperature of 32.5°C and reaction time of 210 ms indicate that cooling had affected physiological function. The player should therefore be allowed sufficient time to rewarm and recover before returning to competition.

Final judgement

Player 1's structured warm-up is the strategy that best balances performance and physiological safety. Player 3's ice treatment can be retained as an injury-management measure but should be followed by adequate recovery before returning to play. Player 2's use of local anaesthetic to continue playing despite injury should be discouraged because it can mask protective pain signals and was associated with the poorest reaction time.

A sound sports-management strategy should therefore combine proper warm-up before activity, appropriate assessment of injuries, controlled use of cold treatment when indicated, adequate recovery time, hydration and avoidance of interventions that simply mask pain to permit continued play.

ITEM 5

(a) Analyse the data to explain how the water hyacinth affected the available fish foods and how the distinct physical changes in fish jaw size came about over the 15-year period.

The data show a clear relationship between the **increase in water hyacinth coverage and the change in the distribution of fish jaw sizes**. In 2009, hyacinth coverage was only **5%**, and the fish population was dominated by intermediate-jawed fish at **80%**, while small-jawed and large-jawed fish accounted for only **10% each**. By 2024, hyacinth coverage had increased to **85%**, while intermediate-jawed fish had declined drastically to only **4%**. At the same time, small-jawed fish increased from **10% to 48%**, and large-jawed fish increased from **10% to 48%**. This indicates that increasing hyacinth coverage changed the feeding environment and created different selection pressures that favoured fish with different jaw sizes.

The increase in **hyacinth coverage from 5% in 2009 to 35% in 2014, 65% in 2019 and 85% in 2024** would have drastically altered light penetration into the lake. **This is linked to changes in the types and availability of fish food in the lake.** Water hyacinth forms thick mats on the water surface, reducing the amount of sunlight reaching the water below. Light is

required by aquatic plants and algae for photosynthesis. Therefore, extensive hyacinth coverage reduces primary production by limiting light available to submerged plants and algae. Since these organisms form the base of many aquatic food chains, their decline can reduce the availability of some food resources for fish.

The thick hyacinth mats would also alter the **physical distribution and accessibility of food organisms**. Some organisms associated with open, well-lit water may decline, while organisms living among or around dense vegetation may become more abundant. Consequently, fish populations would face different feeding opportunities and competition for available food. **This change in food availability provides a biological explanation for why the originally dominant intermediate-jawed fish declined while small- and large-jawed fish increased.**

The **intermediate-jawed fish declined from 80% in 2009 to 50% in 2014, 16% in 2019 and only 4% in 2024**. This dramatic decline is linked to the increasing hyacinth coverage and the associated changes in available food resources. Since the original population had a relatively uniform intermediate jaw size, the fish were likely adapted to exploiting the food resources that were commonly available before extensive hyacinth invasion. As the environment changed, some of the food resources available to these fish became less accessible or less abundant. Intermediate-jawed fish would therefore have experienced greater competition for suitable food, reducing their survival and reproductive success.

In contrast, **small-jawed fish increased from 10% in 2009 to 25% in 2014, 42% in 2019 and 48% in 2024**. **This increase is linked to the changing food environment created by the expanding hyacinth mats.** If the altered ecosystem favoured smaller food particles or organisms associated with the vegetation, fish possessing smaller jaws would have been better able to capture and utilise these resources. Such fish would obtain sufficient food to survive and reproduce, causing the proportion of small-jawed individuals to increase over successive generations.

Similarly, **large-jawed fish increased from 10% in 2009 to 25% in 2014, 42% in 2019 and 48% in 2024**. **This increase indicates that the changing ecosystem also favoured individuals capable of exploiting larger or different food resources that remained available under the new conditions.** Fish with larger jaws may be able to handle larger prey or different food items more effectively. Individuals possessing advantageous jaw characteristics would therefore have a greater probability of obtaining food, surviving and reproducing.

The progressive changes in jaw size can be explained through **natural selection acting on variation within the fish population**. Although the original population was described as having a uniform intermediate jaw size, genetic variation can arise through mutation and recombination. When the environment changed because of increasing hyacinth coverage, different jaw characteristics became more advantageous. Fish that were better suited to the newly available food resources had a greater chance of survival and reproduction. Their advantageous characteristics could be passed to their offspring, causing the frequency of those characteristics to increase in the population over generations.

This is supported by the **gradual rather than sudden change in the data**. Small-jawed fish increased from **10% to 25%, then 42% and finally 48%**, while large-jawed fish followed a similar pattern. At the same time, intermediate-jawed fish progressively decreased from **80%**

to 50%, 16% and finally 4%. The gradual shift over 15 years is consistent with repeated differential survival and reproduction under changing environmental conditions rather than an immediate transformation of individual fish.

The increasing hyacinth coverage also affected other organisms in the ecosystem. The researchers observed that **African eagle populations decreased while moorhen populations increased**. This suggests that the invasion altered food webs and habitat conditions beyond the fish population. Dense water hyacinth can provide habitat and shelter for organisms such as moorhens, while changes in fish populations and aquatic food availability can affect predators such as African eagles. Therefore, the hyacinth invasion produced effects at several trophic levels and contributed to wider ecosystem changes.

Overall, the data indicate that increasing water hyacinth coverage **changed light penetration and the availability and accessibility of fish food**, creating new selection pressures. Fish with jaw characteristics better suited to the altered food resources had greater survival and reproductive success. Over generations, this resulted in an increase in small- and large-jawed fish and a marked decline in intermediate-jawed fish. Thus, the distinct jaw-size distribution observed in 2024 can be explained by **environmental change followed by natural selection and differential reproductive success**.

(b) Propose two more ways to restore the ecosystem and compare them to the effectiveness of that proposed by the researchers and give strategic guidance.

The researchers proposed introducing *Neochetina bruchi* **weevils**, which feed on water hyacinth. This is a form of **biological control** intended to reduce the invasive plant population naturally.

1. Mechanical removal of water hyacinth

One additional strategy would be the **mechanical removal of water hyacinth using boats, harvesters or organised manual removal**. This could be effective because it directly removes the plant from the lake rather than waiting for its population to decline. Reducing the dense mats would allow more light to penetrate the water, potentially restoring photosynthesis by submerged plants and algae. This could improve primary productivity and eventually increase the availability of food resources for fish.

However, mechanical removal can be expensive and labour-intensive, particularly when the infestation covers **85% of the lake**. If the roots or fragments are left behind, the plant can regenerate and rapidly recolonise the water. Removal also needs to be repeated because water hyacinth has a high reproductive capacity.

Compared with the proposed *Neochetina bruchi* biological control, mechanical removal has the advantage of producing a relatively rapid reduction in hyacinth coverage, whereas weevils may require time to establish a population and produce sufficient feeding pressure. **However, the weevils can provide longer-term biological suppression with less continuous mechanical labour once they become established.**

Therefore, mechanical removal would be most useful for **rapidly opening important areas of the lake**, especially fishing grounds, breeding areas and water-access points, while biological control could provide longer-term suppression.

2. Prevention of nutrient enrichment and control of pollution

A second strategy would be to **reduce the amount of nutrients entering Lake Kyoga from agricultural runoff, domestic sewage and other sources**. Excess nutrients such as nitrates and phosphates can promote rapid growth of aquatic plants and algae. Reducing nutrient inputs would therefore limit the conditions that encourage water hyacinth proliferation.

This approach would help address the **underlying environmental conditions that support the invasion**, rather than only removing existing plants. Reduced nutrient loading could also improve water quality and support recovery of native aquatic vegetation and organisms. A healthier aquatic ecosystem would provide more stable food resources for fish and help restore ecological balance.

However, reducing nutrient input would not immediately remove the existing 85% hyacinth coverage. The plants already present would continue growing and reproducing unless combined with another control method. Therefore, this approach is slower in producing visible results than mechanical removal or biological control.

Compared with *Neochetina bruchi*, nutrient management has the important advantage of addressing one of the factors that can sustain excessive aquatic plant growth. However, the weevils directly attack the hyacinth and can therefore reduce the existing infestation more directly. The most effective approach would consequently combine the two rather than relying on either one alone.

Evaluation of the proposed *Neochetina bruchi* strategy

Introducing *Neochetina bruchi* **weevils is potentially effective because the insects feed specifically on water hyacinth.** Their feeding damages the plant and can reduce its growth and reproductive ability. As the hyacinth population decreases, more sunlight can penetrate the water, allowing submerged plants and algae to photosynthesise more effectively. Recovery of these primary producers can increase food availability at lower trophic levels and eventually support recovery of fish populations.

However, biological control may act relatively slowly because the weevil population must establish and reproduce before sufficient feeding pressure is achieved. Its effectiveness may also depend on environmental conditions and the ability of the introduced organisms to survive and spread throughout the affected areas. In addition, biological control must be carefully monitored to ensure that it does not produce unintended ecological effects.

Therefore, *Neochetina bruchi* **is a promising long-term control method, but it should not be relied upon as the only intervention.**

Strategic guidance

The most effective strategy would be an **integrated ecosystem-management approach** rather than choosing only one method. Initially, **mechanical removal** should be used to rapidly reduce the dense hyacinth mats in critical areas. At the same time, carefully monitored introduction of *Neochetina bruchi* could provide longer-term biological suppression of the remaining plants.

In addition, the district should **control nutrient and pollutant inputs into the lake** by improving sewage management, controlling agricultural runoff and establishing vegetation buffers along the shoreline. This would reduce the conditions that encourage continued proliferation of water hyacinth.

The recovery should also be monitored using indicators such as **hyacinth coverage, fish jaw-size distribution, fish population abundance, aquatic plant diversity and bird populations**. A successful intervention should gradually reduce hyacinth coverage, allow intermediate-jawed fish populations and other native organisms to recover where suitable food resources return, and restore balance between species such as African eagles and moorhens.

Final judgement

Neochetina bruchi **introduction is a potentially effective long-term strategy, while mechanical removal provides faster control and nutrient management addresses the underlying conditions favouring the invasion.** Therefore, the best strategic guidance is to **combine the three approaches**: rapidly remove existing mats, use biological control for sustained suppression, and reduce nutrient pollution to prevent renewed proliferation. This integrated approach offers a stronger possibility of restoring the lake's food web, biodiversity and ecosystem stability than relying on biological control alone.

ITEM 6

(a) Using Table 6A, explain the inheritance pattern of heat tolerance in cattle and the selective pressures acting on each genotype. Predict how the frequencies of (TˆH), (TˆM), and (TˆS) alleles would change under prolonged drought conditions, and explain why the (TˆS) allele remains in the population despite strong selection against it. Using Table 6B, describe how varying grazing pressure affected the ecosystem.

Inheritance pattern and selection acting on the cattle genotypes

The cattle possess three alleles, **(T^H), (T^M), and (T^S)**, with (T^H) conferring extreme tolerance and being dominant over (T^M), while both (T^H) and (T^M) are dominant over (T^S), which confers sensitivity. This means that the expression of heat-tolerance traits

depends on which alleles are present in an individual's genotype. The genotypes in Table 6A therefore show different abilities to cope with the extreme drought conditions.

The **(T^H T^H) genotype has the lowest population frequency of 0.05**, but it has the **lowest mean daily water intake of 12 L and the lowest body-mass loss of only 4%**. This indicates that cattle with this genotype are highly efficient at surviving extreme drought conditions. **This is linked to the advantage of extreme heat and water-retention tolerance described in the scenario.** Because these cattle lose relatively little body mass despite consuming the least water, they are more likely to survive when water is scarce. They are therefore under **positive directional selection**, meaning individuals with this genotype have a greater probability of surviving and reproducing than less tolerant cattle.

The **(T^H T^S) genotype occurs at a frequency of 0.15 and has a mean daily water intake of 14 L with only 6% body-mass loss**. Its relatively low body-mass loss compared with the (T^M T^M), (T^M T^S) and (T^S T^S) genotypes indicates that the presence of the dominant (T^H) allele provides a strong survival advantage under drought. **This is because (T^H) is dominant over (T^S), so the extreme-tolerance phenotype is expressed despite the presence of the sensitive allele.** Consequently, cattle carrying (T^H) are likely to survive and reproduce more successfully during prolonged drought.

The **(T^M T^M) genotype has a frequency of 0.30, consumes 25 L of water per day and loses 15% of body mass**. Compared with (T^H T^H) cattle, these animals require considerably more water and lose more body mass. **This indicates that they have moderate rather than extreme tolerance and therefore experience stronger selection pressure during severe drought.** They can survive better than sensitive cattle but are less well adapted than cattle carrying (T^H). Thus, prolonged drought would favour (T^H) over (T^M), although (T^M) would not necessarily be eliminated because it still provides greater tolerance than (T^S).

The **(T^M T^S) genotype has a frequency of 0.40, the highest of all the genotypes shown, and has a mean daily water intake of 28 L with 19% body-mass loss**. Its greater water requirement and body-mass loss indicate that these cattle experience considerable physiological stress during drought. **This is linked to the moderate-tolerance phenotype associated with (T^M), while the (T^S) allele does not provide additional tolerance.** Therefore, these cattle are under stronger negative selection than (T^H)-carrying cattle because they are less able to conserve water and maintain body condition.

The **(T^S T^S) genotype has a frequency of 0.10 and shows the poorest survival characteristics**, requiring the greatest daily water intake of **45 L** and losing the greatest proportion of body mass, **35%**. **This is linked to the sensitive phenotype produced when an individual has two (T^S) alleles.** Under prolonged drought, these cattle would be at the greatest disadvantage because their high water requirement cannot easily be met in a semi-arid environment. Their large body-mass loss reduces their physiological condition and potentially their ability to reproduce. Therefore, (T^S T^S) cattle experience the strongest **negative selection**.

Predicted changes in allele frequencies

Under prolonged drought, the **(T^H) allele would be expected to increase in frequency** because cattle carrying it have the lowest water requirement and lowest body-mass loss.

These individuals are more likely to survive the drought and reproduce successfully, passing (T^H) to their offspring. Over successive generations, natural selection would therefore increase the representation of (T^H) in the population.

The **(T^M) allele would probably decrease in frequency**, because cattle with (T^M) require considerably more water and lose more body mass than cattle carrying (T^H). However, its decline would probably be slower than that of (T^S), because (T^M) still provides moderate tolerance and therefore offers some survival advantage during drought.

The **(T^S) allele would be expected to decrease under prolonged drought**, because cattle expressing the sensitive phenotype suffer the greatest body-mass loss. The (T^S T^S) genotype loses **35% of body mass**, compared with only **4% in (T^H T^H)** cattle. Consequently, sensitive cattle would have lower survival and reproductive success, reducing transmission of (T^S) to subsequent generations.

Why the (T^S) allele remains in the population

Despite strong selection against (T^S), the allele would **not necessarily disappear immediately** because it can remain hidden in heterozygous individuals. Since both (T^H) and (T^M) are dominant over (T^S), an individual with (T^H T^S) or (T^M T^S) does not necessarily express the sensitive phenotype in the same way as (T^S T^S).

This is important because the table contains **0.15 (T^H T^S)** and **0.40 (T^M T^S)** individuals. These heterozygotes can survive and reproduce, allowing them to pass the (T^S) allele to their offspring even though they may not show the severe sensitivity associated with (T^S T^S). Therefore, **selection acts most strongly against the phenotype expressed by (T^S T^S), while (T^S) can remain hidden in heterozygotes**.

Consequently, prolonged drought would gradually reduce the frequency of (T^S), but it would not necessarily eliminate it completely. The allele can persist in carriers and can reappear as (T^S T^S) when two carriers reproduce.

Effect of varying grazing pressure on the ecosystem

Table 6B shows a clear deterioration of ecosystem condition as grazing pressure increased.

Under **light grazing**, *Themeda triandra*, a nutritious perennial grass, was the dominant plant species. Net primary productivity was **500 g/m²/yr**, while soil erosion was only **0.5 tons/ha/yr**. **This indicates that light grazing allowed sufficient vegetation to remain on the land to maintain high primary productivity and protect the soil surface.** The extensive root systems of perennial grasses help bind soil particles, while above-ground vegetation reduces the direct impact of rainfall on exposed soil. Consequently, erosion remained very low.

Under **moderate grazing**, the vegetation changed to **mixed grasses and herbs**, while net primary productivity fell to **200 g/m²/yr** and soil erosion increased sharply to **4.2 tons/ha/yr**.

This decline is linked to increased removal of plant biomass by cattle. Greater grazing pressure reduces the amount of leaf area available for photosynthesis, reducing the amount of organic matter produced by plants. At the same time, repeated grazing can reduce vegetation cover, leaving more soil exposed to wind and water erosion. Therefore, moderate grazing reduced ecosystem productivity and increased soil degradation compared with light grazing.

Under **heavy overgrazing**, the dominant plant changed to *Sporobolus pyramidalis*, described as an invasive, unpalatable weed. Net primary productivity fell further to only **110 g/m²/yr**, while soil erosion increased dramatically to **38.0 tons/ha/yr**. **This is linked to severe removal of nutritious perennial grasses by excessive grazing.** Cattle preferentially consume palatable grasses such as *Themeda triandra*, giving less-palatable or invasive species a competitive advantage because they are not consumed as heavily. Over time, the nutritious grasses decline while unpalatable weeds become dominant.

The dramatic increase in soil erosion from **0.5 tons/ha/yr under light grazing to 38.0 tons/ha/yr under heavy overgrazing** demonstrates that excessive grazing severely reduces the ecosystem's ability to protect the soil. Removal of vegetation exposes the soil surface, while loss of roots reduces soil binding. Rainfall can therefore detach and transport soil particles more easily, resulting in severe erosion.

Overall, increasing grazing pressure produced a **negative ecological succession**, characterised by replacement of productive perennial grasses with less-palatable invasive vegetation, reduction in primary productivity and increased soil erosion. Therefore, the data demonstrate that uncontrolled grazing threatens both the productivity and long-term sustainability of the rangeland.

(b) Evaluate the long-term sustainability and effectiveness of the ecologists' implementation strategies.

The ecologists propose **mass culling of cattle showing traits of the (T^S) genotype** and the use of **broad-spectrum herbicides to remove** *Sporobolus pyramidalis* so that native grasses can recover. These strategies may produce some short-term benefits, but their long-term sustainability requires careful evaluation.

1. Mass culling of cattle showing (TˆS) traits

Mass culling could be effective in the short term because removing cattle that are poorly adapted to extreme drought could reduce the number of sensitive individuals in the population. The (T^S T^S) genotype has the **highest water requirement of 45 L/day and the greatest body-mass loss of 35%**. Removing these highly sensitive cattle could reduce livestock losses during severe drought and reduce pressure on limited water resources.

Culling could also help reduce grazing pressure if the overall cattle population is too large. The data show that **heavy overgrazing reduces primary productivity from 500 g/m²/yr under light grazing to only 110 g/m²/yr and increases soil erosion from 0.5 to 38.0 tons/ha/yr**. Reducing cattle numbers could therefore allow *Themeda triandra* and other nutritious grasses to recover, increasing vegetation cover and reducing erosion.

However, mass culling is not necessarily a sustainable genetic-management strategy. The presence of (T^S) in heterozygotes means that simply removing animals showing sensitive traits may not eliminate the (T^S) allele from the population. Because (T^S) can be carried by (T^H T^S) and (T^M T^S) individuals without the same severe phenotype, the allele may continue to be transmitted to offspring.

Furthermore, indiscriminate mass culling could reduce the **genetic diversity of the cattle population**. If animals carrying useful combinations of alleles are removed unnecessarily, the population may lose genetic resources that could be valuable under future environmental conditions. It could also reduce herd sizes beyond sustainable levels and negatively affect pastoralists who depend on cattle for livelihoods.

Therefore, mass culling should not be the primary long-term strategy. A more sustainable approach would be **selective breeding for drought-tolerant cattle**, particularly increasing the representation of (T^H) while maintaining useful genetic diversity. At the same time, stocking rates should be controlled according to the carrying capacity of the rangeland.

2. Use of broad-spectrum herbicides against *Sporobolus pyramidalis*

The use of herbicides could produce a **rapid reduction in the invasive weed**, allowing native nutritious grasses such as *Themeda triandra* to recolonise the rangeland. This could improve vegetation cover and increase primary productivity. Since heavy overgrazing is associated with only **110 g/m²/yr** of productivity, removing the invasive competitor could create space and resources for more productive native vegetation.

However, broad-spectrum herbicides do not selectively distinguish between the invasive weed and desirable native plants. If applied extensively, they may also kill native grasses and other vegetation that the intervention is intended to restore. This could temporarily increase the amount of bare soil and consequently worsen erosion. This is especially concerning because soil erosion is already extremely high under heavy overgrazing, reaching **38.0 tons/ha/yr**.

Broad-spectrum herbicides may also affect other components of the ecosystem, including non-target plants and organisms associated with the vegetation. Removing plant species without addressing the cause of their dominance may also allow the weed to recolonise once the herbicide effect decreases.

Therefore, broad-spectrum herbicide use may be effective for rapid weed suppression but is not ideal as a stand-alone long-term conservation strategy.

A more sustainable approach would involve **targeted weed control**, combined with restoration of native grasses and proper grazing management. Herbicide should be applied selectively where necessary rather than over large areas.

Strategic guidance

The strongest long-term strategy would be to **combine controlled livestock management, genetic improvement and targeted vegetation restoration rather than relying on mass culling and broad-spectrum herbicides alone**.

First, cattle numbers should be adjusted to the **carrying capacity of the rangeland**. Rotational grazing can allow sections of pasture to recover while preventing continuous grazing of the same vegetation. This would help restore *Themeda triandra*, increase primary productivity and reduce soil erosion.

Second, instead of mass culling, the pastoralists should adopt a **selective breeding programme favouring cattle carrying the (T^H) allele**, because (T^H)-carrying cattle show the best performance under drought. The (T^H T^H) cattle lose only **4% body mass**, compared with **35% in (T^S T^S)** cattle. Breeding from the better-adapted animals would gradually increase drought tolerance while avoiding unnecessary loss of genetic diversity.

Third, management of *Sporobolus pyramidalis* should involve **targeted rather than broad-spectrum herbicide application**, followed by reseeding or protection of native perennial grasses. Once nutritious grasses are re-established, controlled grazing can help maintain a productive plant community.

Finally, the rangeland should be monitored using indicators such as **cattle genotype frequencies, body-mass loss, vegetation composition, primary productivity and soil erosion**. This would allow managers to determine whether their interventions are actually improving ecosystem condition.

Overall judgement

Mass culling and broad-spectrum herbicide use may provide short-term benefits, but neither is sufficiently sustainable as a stand-alone long-term strategy. Mass culling can reduce grazing pressure and remove highly drought-sensitive cattle, but it may reduce genetic diversity and will not necessarily eliminate the recessive (T^S) allele because it can remain hidden in heterozygotes. Broad-spectrum herbicides can rapidly suppress *Sporobolus pyramidalis*, but they may also damage desirable native vegetation and leave soil exposed to erosion.

Therefore, the recommended strategy is an integrated approach: reduce excessive stocking, use rotational grazing, selectively breed drought-tolerant cattle carrying (T^H), apply targeted weed control, restore *Themeda triandra* and other native grasses, and continuously monitor vegetation and soil condition. This approach addresses both the **genetic problem in the cattle population and the ecological problem caused by overgrazing**, making it more sustainable for the semi-arid rangelands of Kotido District.

END